Lecture 22
Sample Size and Chance

Review: which interval?

a proportion p

$\hat{p} \pm z \sqrt{\hat{p}(1 - \hat{p}) / n}$, when $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$

a mean μ, σ known

$\bar{x} \pm z \sigma / \sqrt{n}$

a mean μ, σ unknown

$\bar{x} \pm t s / \sqrt{n}$, with $df = n - 1$

when the t interval works

the population is roughly normal, or $n \geq 30$

Data: seal whiskers

Ginter et al. (2012) measured the longest undamaged whisker of 92 seals and sea lions from 11 species, to study how whisker shape evolved.

X = the total length of a seal's longest whisker, in mm

SPECIES	SEALS (n)	MEAN LENGTH (mm)	SD (mm)
Harp seal	16	80.6	10.8
Harbor seal	15	86.5	10.5
Ringed seal	16	79.6	10.9
Spotted seal	9	91.2	7.9
Gray seal	8	73.8	12.5
…	…	…	…

One whisker per seal; only each species' n, mean and SD were published

Example: spotted seal whiskers

Estimate μ, the mean whisker length of all spotted seals, from the 9 that were measured.

n = 9; $\bar{x}$ = 91.2 mm; s = 7.9 mm

(a) z or t? How many degrees of freedom?
(b) What is the standard error?
(c) What is the 95% confidence interval for μ?

(a) t, because σ is unknown; df = 9 – 1 = 8, t = 2.306
(b) 7.9 / √9 ≈ 2.63 mm
(c) 91.2 ± 2.306(2.63) = 91.2 ± 6.06: 85.1 to 97.3 mm

Choosing a sample size

margin of error m

the ± part of an interval: $m = z \times SE$, so a larger n gives a smaller m

n for a proportion

$n = z^2 p^*(1 - p^*) / m^2$, with $p^* = 0.5$ if there is no earlier guess

n for a mean

$n = (z \sigma / m)^2$, with σ from an earlier study

always round up

n = 59.1 means 60: rounding down misses the target m

Example: planning the next study

Before collecting data, decide how many to measure so the 95% interval is as narrow as you need.

carriers: m = 0.03, p* = 0.5; whiskers: m = 2 mm, σ ≈ 7.9 mm

(a) How many people for the proportion of *2 carriers?
(b) How many spotted seals for the mean whisker length?
(c) What happens to n if we want half the margin of error?

(a) $1.96^2\,(0.5)(0.5) / 0.03^2 = 1067.1$ → 1,068 people
(b) $(1.96 \times 7.9 / 2)^2 = 59.9$ → 60 seals
(c) 4 times as large: m is squared on the bottom

Practice: how many students?

Campus dining wants to estimate p, the proportion of students who skip breakfast, to within 4 percentage points with 90% confidence.

m = 0.04; 90% confidence; last year's survey: 30% skip breakfast

(a) How many students should they survey, using last year's 30%?
(b) How many if there were no earlier survey?
(c) Why is (b) larger?

(a) $1.645^2\,(0.3)(0.7) / 0.04^2 = 355.2 \rightarrow$ 356 students
(b) $1.645^2\,(0.5)(0.5) / 0.04^2 = 422.8 \rightarrow$ 423 students
(c) $p^*(1 - p^*)$ is largest at $p^* = 0.5$, so 0.5 is the safe guess

Can you taste the difference?

Your roommate claims they can tell Coke from Pepsi. In a blind test, you pour one of the two at random, 8 times, and they name each cup.

8 of 8 cups named correctly

(1) Can they really tell, or were they just lucky? How could we decide?
(2) If they were only guessing, how likely is 8 for 8?
(3) What are the two ways to explain such a small probability?

(2) $(1/2)^8 = 1/256 \approx 0.004$
(3) Either something rare happened by chance, or "just guessing" is wrong
How rare is rare enough is our choice; by convention, below 0.05